Missing and Unbalanced Examples

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Missing and Unbalanced Examples

> Prepare Missing and Unbalanced Examples: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Name the target collection, selection list and cases actually assessed

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Good Data, Bad Data

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

Look for: Check records against a stated format, unit and meaning before using them

Reference example, not the only acceptable wording: Height 60 fails the whole-number range 1 through 10. Travel blank is missing; its value cannot be inferred.

Travel 0 is valid in the whole-number range 0 through 10 and can represent observed no travel. Zero is a measurement, not a replacement for a blank.

Similarity

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate similarity using explicitly selected features and a stated scoring rule

Reference example, not the only acceptable wording: A: 1, 1, 0 gives 2. B: 0, 0, 1 gives 1. C: 1, 0, 0 gives 1. A is the only highest scoring candidate among A, B and C under equal weights.

Boundaries and 'Not Sure'

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a specific source of uncertainty and explain how it affects a **prediction**

Reference example, not the only acceptable wording: The rule and two stated possibilities are known; the actual height is unresolved. At 5 predict short, at 6 predict tall. Inspect the original readable record or remeasure.

Repeating a calculation supplies no new information about which height was recorded.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A **training** set has 18 daylight photos and two dark photos of one object. Copying the dark photos eight times balances the counts. Does it balance the variety? Design a better collection step.

Concept to explain

What is absent can shape a system's failures.

Term	Meaning
missing example	needed condition absent from data
imbalance	some groups appear much more than others
subgroup	defined part of the intended population
coverage gap	important situation poorly represented

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach the difference between accurate records and adequate **coverage**. The populations here are finite invented card and clip collections, so students can account for every selected and unselected case.

Avoid presenting equal group counts as a universal sampling rule: this target has equal groups, and the proposed balanced plan investigates condition **coverage**. It does not guarantee an unbiased estimate in every application.

Readiness answer

No. Correct records can still describe only a narrow set of conditions. Source checking does not establish **coverage** of unrecorded cases.

A paper inventory with 6 IDs for each of the four picture-card conditions

A four-row **coverage** table with target, selected, assessed and missing columns

Twenty paper clip IDs grouped into the four fictional sound conditions; no audio recording needed

End goal for the pictorial case

By the end, I can identify missing conditions and design a targeted **coverage** check.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Missing and Unbalanced Examples using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT example set with represented and absent conditions

2. Process — the deciding step

PROCESS learn patterns from available **coverage**

3. Output — the result

OUTPUT uneven performance across conditions

Key idea

CORE IDEA What is absent can shape a system's failures.

How the mechanism works

UNDER THE HOOD Sampling **imbalance** changes the empirical distribution seen during **training**. Missing **subgroup**s or conditions cannot contribute error signals, so average success may hide systematic failure.

Rebalancing requires relevant examples, not arbitrary duplication alone.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE An animal recognizer trained only on bright daytime photos may fail on the same animals at night.

Step 1

STEP 1 — OBSERVE example set with represented and absent conditions

Step 2

STEP 2 — TRACE learn patterns from available **coverage**

Step 3

STEP 3 — CONCLUDE uneven performance across conditions

Model answer

MODEL REASONING The model may rely on bright backgrounds or fail to detect shapes in low light. Collect varied night examples for every relevant animal and preserve a separate night test set to evaluate the repair.

A claim to avoid

DO NOT OVERCLAIM Balance is about useful representation, not forcing every count to be identical without reason.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

Training includes nine daylight leaf photos and one low-light photo.

What intended-use gap is present?

Exact explanation to model

Why the result follows

Low light is poorly represented. Add varied, correctly labelled low-light examples if that use is intended. Copies of one photo do not add equivalent variety.

Explain the added worked example

Explain the added worked example · 1

Suppose a shape recogniser will be evaluated on a fixed collection of 24 invented picture cards. This **target population** has 6 bright circles, 6 dim circles, 6 bright squares and 6 dim squares.

Each card has its own ID and an independently checked circle or square label. We want to know how often the recogniser gives the correct label across these 24 cards.

Explain the added worked example · 2

A tester takes the 6 bright-circle cards nearest the front of the box. Those 6 cards form the **sample**; the other 18 have not been assessed. The recogniser is correct on 5 of the 6 **sample**d cards.

That is a result for these assessed bright circles. It does not establish correctness on the squares or dim cards.

Explain the added worked example · 3

A **sampling frame** is the list or source available for selection. A complete list of all 24 IDs could give every card a chance of selection.

A list containing only bright-circle IDs cannot select dim cards or squares, even if the choice from that list is random. These fictional picture cards do not contain personal information.

Explain the added worked example · 4

The target has 24 distinct cards, but the **sample** covers only the 6 bright circles. Photographing each **sample**d card twice would create 12 image records without adding a new card or covering a missing condition.

Record the identity of the underlying case as well as the number of files.

Explain the added worked example · 5

Both shape and lighting matter to this question. A **sample** with bright circles and dim squares includes each shape and each lighting label, yet still misses dim circles and bright squares.

Check combinations when the interaction could matter, not just whether each individual label appears somewhere.

Key terms

Term	Meaning
Target population	The whole collection of cases a question is about
Sample	The selected cases used to investigate a larger collection
Sampling frame	The list or source from which cases can be selected
Coverage	Which relevant kinds of cases the list or sample includes
Selection bias	Systematic distortion caused by how cases enter an investigation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Plan	What it helps with	Limit to report
Take cards at the front	Quick initial check	Front position may restrict conditions
Draw from all 24 IDs by chance	Every listed card can be selected	A small draw need not balance conditions
Choose 2 distinct cards per condition	Includes all four conditions	8 cards do not establish all 24 outcomes
Assess all 24 distinct cards	Covers this entire fixed collection	Does not prove performance on future images

Model the reasoning aloud · 6

For a broad estimate, a chance-based plan should start from a suitable complete list and use a documented selection procedure.

For investigating missing conditions, deliberately selecting cases from each condition can be useful. State which purpose you chose. A set built to expose problems is not automatically a faithful miniature of everyday use.

Model the reasoning aloud · 7

Selection bias is about a systematic feature of the selection **process**, not whether a person had bad intentions. Choosing only easy-to-reach, bright cards can leave harder conditions out.

More cases from the same narrow source may repeat that gap. A larger **sample** cannot by itself repair a frame that excludes relevant cases.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Missing and Unbalanced Examples

A fictional badge reader will be used in both bright and dim rooms.

The facts we will use

> **Supplied facts**
>
> Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards.

> **Supplied facts**
>
> A development **sample** contains 6 bright plain cards and 2 bright patterned cards; no dim

cards.

AI ACADEMY / AI-1/22

Target groups and development coverage

	Target cards	Sample cards
Bright plain	8	6
Bright patterned	8	2
Dim plain	8	0
Dim patterned	8	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/22

Missing and Unbalanced Examples

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Target conditions

Four lighting-by-pattern groups, eight target cards each.



2. Current sample

Eight bright cards only; dim groups have zero coverage.



3. Coverage gap

Current results cannot describe dim-room reading.



4. Targeted follow-up

Select and record new cases from both dim groups before claiming coverage.

CHANGE ONE THING

Add the two remaining bright plain cards and no dim cards.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Target conditions	Four lighting-by-pattern groups, eight target cards each.
2. Current sample	Eight bright cards only; dim groups have zero coverage.
3. Coverage gap	Current results cannot describe dim-room reading.
4. Targeted follow-up	Select and record new cases from both dim groups before claiming coverage.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

What this example does not establish

Keep the stated limit

Coverage does not guarantee accuracy, and targeted cases alone do not supply a population-wide success estimate.

Change one thing in the pictured case

Change one thing

Change: Add the two remaining bright plain cards and no dim cards.

Predicted result

Prediction: The **sample** becomes larger but still has no dim **coverage**.

Why the result follows

Reason: The missing lighting condition remains untested.

Ask while showing the picture

Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Missing and Unbalanced Examples, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: **Training** includes nine daylight leaf photos and one low-light photo.

2. State the decision to explain

Write the question to explain: What intended-use gap is present?

3. Reveal the worked reasoning

Reveal after **prediction**s: Low light is poorly represented. Add varied, correctly labelled low-light examples if that use is intended. Copies of one photo do not add equivalent variety.

Hold back the independent task

Keep for an independent attempt: A photo sorter trains on ten bright photos and no dim photos. What group of examples is missing? Design a test that reveals the gap.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The original recogniser correctly labelled 5 of 6 assessed bright-circle cards. Dim circles, bright squares and dim squares were not assessed.

A documented plan selecting 2 distinct cards per condition from all 24 IDs would include each condition without claiming to establish every outcome.

Exit: the image count grew, but the absence of dim-square tests remains; more bright-circle data cannot supply dim-square **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve why source-checked data can still omit relevant conditions.
7 to 20 minutes	Model target, frame and sample using the picture-card inventory. Complete W1.
20 to 33 minutes	Read the coverage diagram. Complete W2 and distinguish file count from distinct cases.
33 to 45 minutes	Complete W3 and compare individual features with their combinations.
Session 2 start 0 to 8 minutes	Retrieve the three absent conditions in the original investigation.
8 to 23 minutes	Compare selection plans and complete the guided claim rewrite. Use shuffled paper IDs to show that a small chance draw need not balance conditions.

Stage	Teacher action
23 to 38 minutes	Collect W4 independently. Require a bounded result and a feasible plan with distinct clips.
38 to 45 minutes	Discuss W5 and the exit. Keep selected, assessed and unavailable counts separate.

Reduce help deliberately

Completion practice

Completion task: Each of the six original **sample**d cards is photographed twice. State the number of image files and distinct cards. Has this covered a new shape or lighting condition?

Explain why the counts need different labels.

Reasoning to listen for

Teacher check: There are 12 image files but still 6 distinct cards, all bright circles. No new shape or lighting condition has been covered.

Multiple records of the same cases are not 12 distinct **sample**d cards and do not resolve the missing conditions.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Sampling and **coverage**. Proposed explanation: “More records always remove bias”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A restricted source can keep excluding the same conditions as its record count grows.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Would testing only dim plain cards close every current gap? Point to the target groups.

Reasoned teacher answer

No. Dim patterned cards would still be absent; both dim groups need planned **coverage**.

Changed-case reasoning: The **sample** becomes larger but still has no dim **coverage**. The missing lighting condition remains untested.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Missing and Unbalanced Examples investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Predict failures in an animal set missing night photos.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

> **Find the unsupported claim**
>
> CLAIM TO REPAIR Ten examples cover every condition.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

More records always remove bias: A restricted source can keep excluding the same conditions as its record count grows.

Random selection repairs an incomplete list: Excluded cases cannot be selected from a list that does not contain them.

Every feature value appears, so every combination appears: Bright circles and dim squares still omit dim circles and bright squares.

Zero assessed means every case failed: Zero assessed means no observed recognition outcomes for that group.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Missing and Unbalanced Examples questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Low light is poorly represented. Add varied, correctly labelled low-light examples if that use is intended. Copies of one photo do not add equivalent variety.

Independent question

Independent question · withhold the key

A photo sorter trains on ten bright photos and no dim photos. What group of examples is missing? Design a test that reveals the gap.

Expected independent answer

Teacher key · Expected independent answer

Dim photos are missing. Test on separately collected, correctly labelled dim photos and compare errors with bright photos. Success in bright light does not establish performance in dim light.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Ten examples cover every condition.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

No. Copies add no new dark situations. Collect permitted photos of several distinct objects under relevant low-light conditions, with reliable labels.

Keep related views together when separating evaluation examples so familiar objects do not inflate results.

Explanations for the additional workbook examples

W1 Name target, frame and sample

Workbook question · W1

Use the 24-card investigation. State the target collection, the original six-card **sample** and the three absent conditions.

If the selection list contains only bright-circle IDs, explain why random selection from that list cannot reach every target condition.

Teacher reasoning · W1

The target is all 24 specified cards across four conditions. The **sample** is the 6 bright-circle cards. Dim circles, bright squares and dim squares are absent.

A frame containing only bright-circle IDs excludes every other condition, regardless of how randomly IDs are drawn from it.

W2 Separate files from distinct cases

Workbook question · W2

Each of the six original **sample**d cards is photographed twice. State the number of image files and distinct cards. Has this covered a new shape or lighting condition? Explain why the counts need different labels.

Teacher reasoning · W2

There are 12 image files but still 6 distinct cards, all bright circles. No new shape or lighting condition has been covered.

Multiple records of the same cases are not 12 distinct **sample**d cards and do not resolve the missing conditions.

W3 Read a coverage table in combinations

Workbook question · W3

A different test includes 3 bright circles and 3 dim squares. A designer says it covers both shapes and both lighting conditions, so no combinations are missing.

Identify the missing combinations and propose six additional distinct cards to investigate them.

Teacher reasoning · W3

Dim circles and bright squares are missing, even though circle, square, bright and dim each occur somewhere. Add 3 distinct dim circles and 3 distinct bright squares from the target collection.

This addresses those combination gaps but does not establish every unassessed card outcome.

W4 Audit a new sound-recognition claim

Workbook question · W4

A fixed target has 20 invented clips: 5 quiet bells, 5 noisy bells, 5 quiet claps and 5 noisy claps. A tester selects 4 distinct quiet bells from a quiet-bell-only list; 3 are correctly labelled and 1 is wrong.

Write a supported result, name missing conditions and explain why the result is not established for all 20 clips. Propose an 8-clip plan that includes every condition, with no clip selected twice.

Teacher reasoning · W4

The recogniser is correct on 3 of 4 assessed quiet-bell clips. No noisy bells, quiet claps or noisy claps were assessed; even one quiet-bell clip remains unassessed.

The restricted frame and **sample** cannot establish the all-20 result. Select 2 distinct clips from each of the four conditions using the complete target list, document selection before outcomes and report remaining limits.

Eight clips improve condition **coverage** but are not all 20.

W5 Report a selected case without an outcome

Workbook question · W5

An 8-clip plan selects 2 from each sound condition. One selected noisy-clap file cannot be opened. The task measures recognition correctness only. State selected and assessed counts and the assessed count for each condition.

Explain why a report of all 8 checked is wrong, and how to handle the missing file transparently.

Teacher reasoning · W5

Eight clips were selected; seven were assessed. Quiet bells, noisy bells and quiet claps each have 2 assessed; noisy claps have 1 assessed and 1 unassessed file. No recognition outcome is available for the unopened file.

Log its ID and reason, recover it if possible, and document any replacement rule without choosing on outcomes. Do not label its recognition result correct or wrong or hide its absence.

Assess independent transfer

Independent question · withhold the key

Selected case: A fixed target has 20 invented clips: 5 quiet bells, 5 noisy bells, 5 quiet claps and 5 noisy claps. A tester selects 4 distinct quiet bells from a quiet-bell-only list; 3 are correctly labelled and 1 is wrong.

Write a supported result, name missing conditions and explain why the result is not established for all 20 clips. Propose an 8-clip plan that includes every condition, with no clip selected twice.

Selected case answer

Teacher key · Selected case answer

The recogniser is correct on 3 of 4 assessed quiet-bell clips. No noisy bells, quiet claps or noisy claps were assessed; even one quiet-bell clip remains unassessed.

The restricted frame and **sample** cannot establish the all-20 result. Select 2 distinct clips from each of the four conditions using the complete target list, document selection before outcomes and report remaining limits.

Eight clips improve condition **coverage** but are not all 20.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Missing and Unbalanced Examples to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

List normal, low-light, partly hidden, worn, and unusual items; mark which conditions are missing from the sorter data.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 23: Fair for Whom?. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Place paper IDs in four labelled boxes and physically mark selected and assessed cards. Provide the column headings while students fill the counts.

Extension: explain why a full assessment of this fixed 24-card population still would not prove correctness on every future photograph.

W4 requires 3 of 4 quiet bells, all three missing conditions, recognition that the all-20 result is unestablished, and an 8-clip plan with distinct cases from all four conditions; 2 per condition is one valid plan.

After support use a **sample** of 4 noisy claps with 2 correct: the supported result is 2 of 4 noisy claps, and quiet bells, noisy bells and quiet claps are absent. Require a fresh explanation, not a copied list.

Week 23 compares performance fractions while keeping denominators, missing outcomes and evaluation conditions explicit.

Plan the delayed return

Suggested return points in this level: ai-1/23, ai-1/25, ai-1/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Missing and Unbalanced Examples

lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Name the target collection, selection list and cases actually assessed

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Each of the six original **sample**d cards is photographed twice. State the number of image files and distinct cards. Has this covered a new shape or lighting condition?

Explain why the counts need different labels.

Supported practice answer

Reasoned practice answer: There are 12 image files but still 6 distinct cards, all bright circles. No new shape or lighting condition has been covered.

Multiple records of the same cases are not 12 distinct **sample**d cards and do not resolve the missing conditions.

Check every learner before discussion: Use the worked case for Sampling and **coverage**. Proposed explanation: “More records always remove bias”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A restricted source can keep excluding the same conditions as its record count grows.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Add the two remaining bright plain cards and no dim cards. Require a **prediction** and a reason before comparing. Expected result: The **sample** becomes larger but still has no dim **coverage**.

Reason: The missing lighting condition remains untested.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

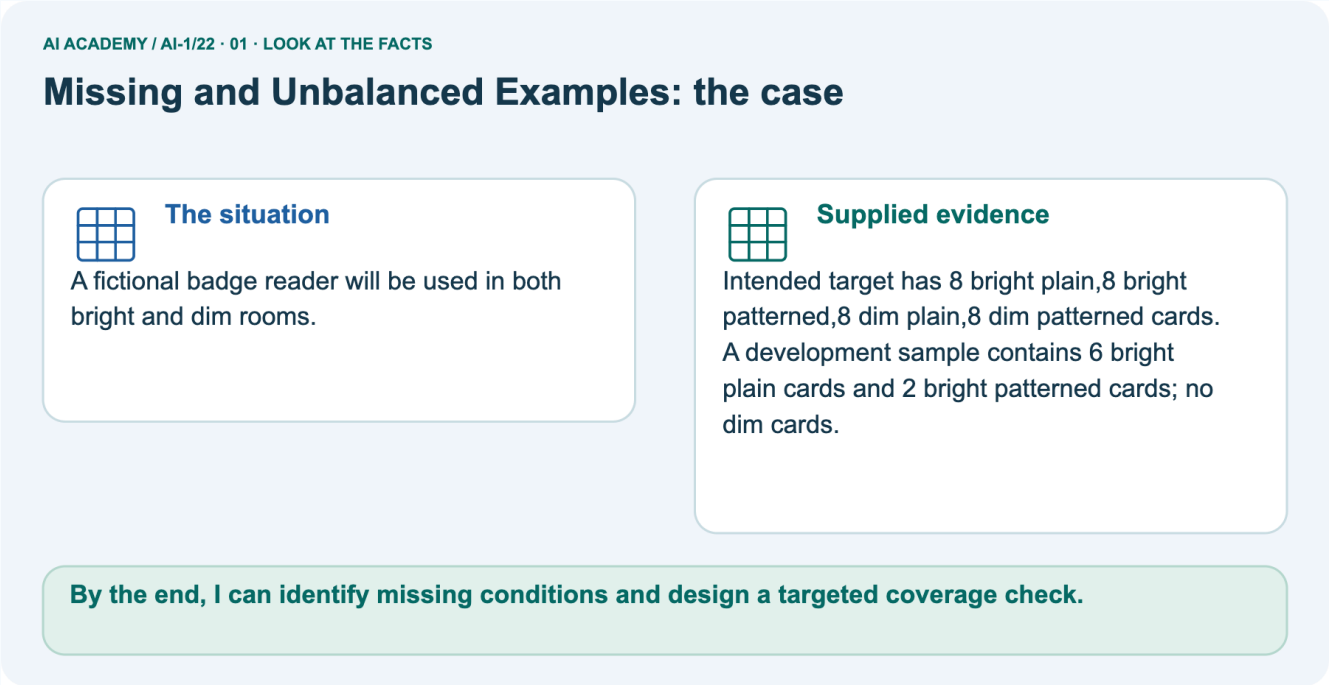
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Visual concept journey

> Show one Missing and Unbalanced Examples graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Missing and Unbalanced Examples: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional badge reader will be used in both bright and dim rooms. Supplied facts: Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards.

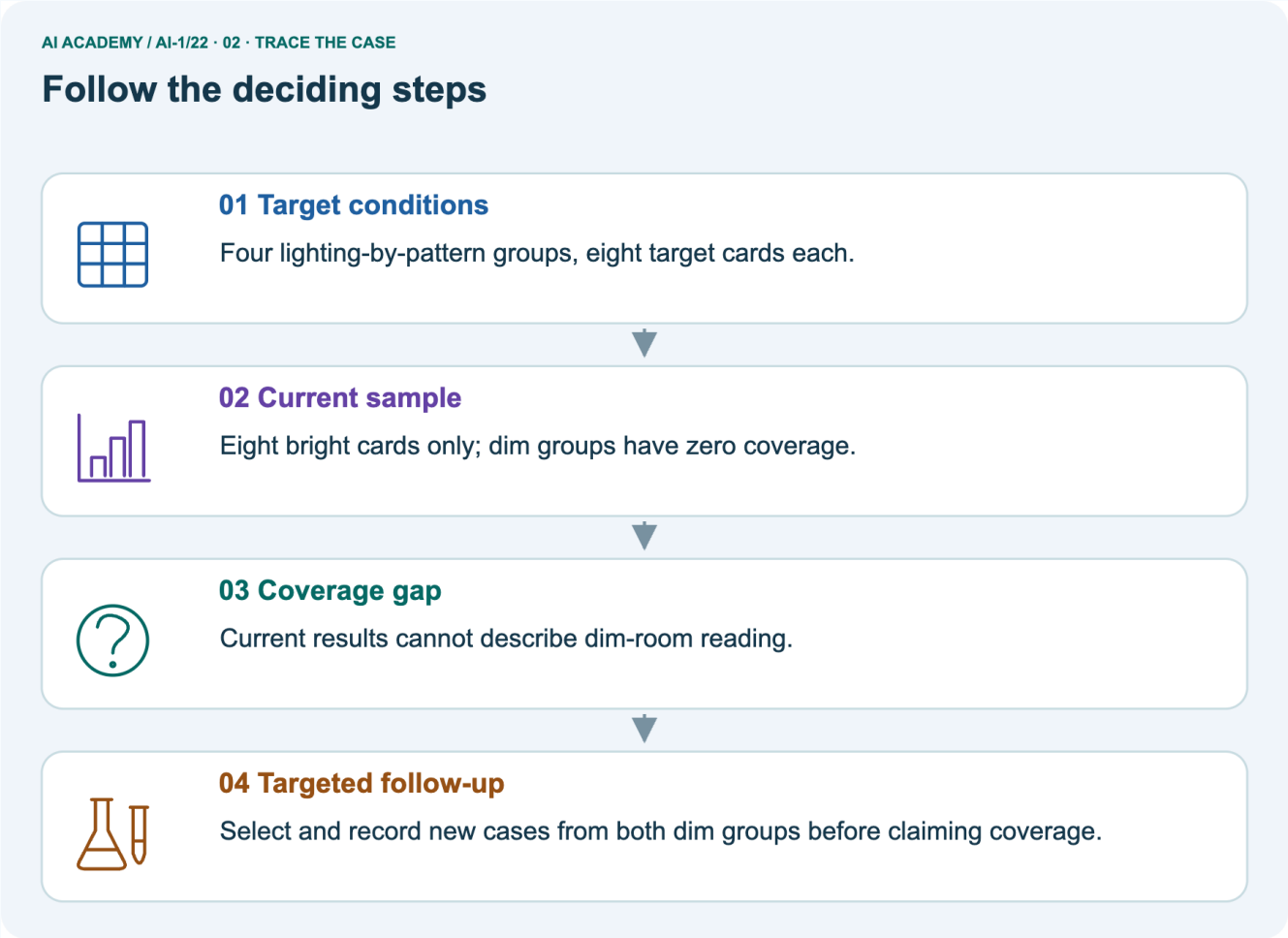
A development **sample** contains 6 bright plain cards and 2 bright patterned cards; no dim cards.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Target conditions: Four lighting-by-pattern groups, eight target cards each. → Current **sample**: Eight bright cards only; dim groups have zero **coverage**. → **Coverage gap**: Current results cannot describe dim-room reading. → Targeted follow-up: Select and record new cases from both dim groups before claiming **coverage**.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Target groups and development coverage



Follow the labelled picture with the matching explanation. Open the vector diagram

Target groups and development **coverage**. Bright plain: Target cards = 8, **Sample** cards = 6. Bright patterned: Target cards = 8, **Sample** cards = 2. Dim plain: Target cards = 8, **Sample** cards = 0.

Dim patterned: Target cards = 8, **Sample** cards = 0.

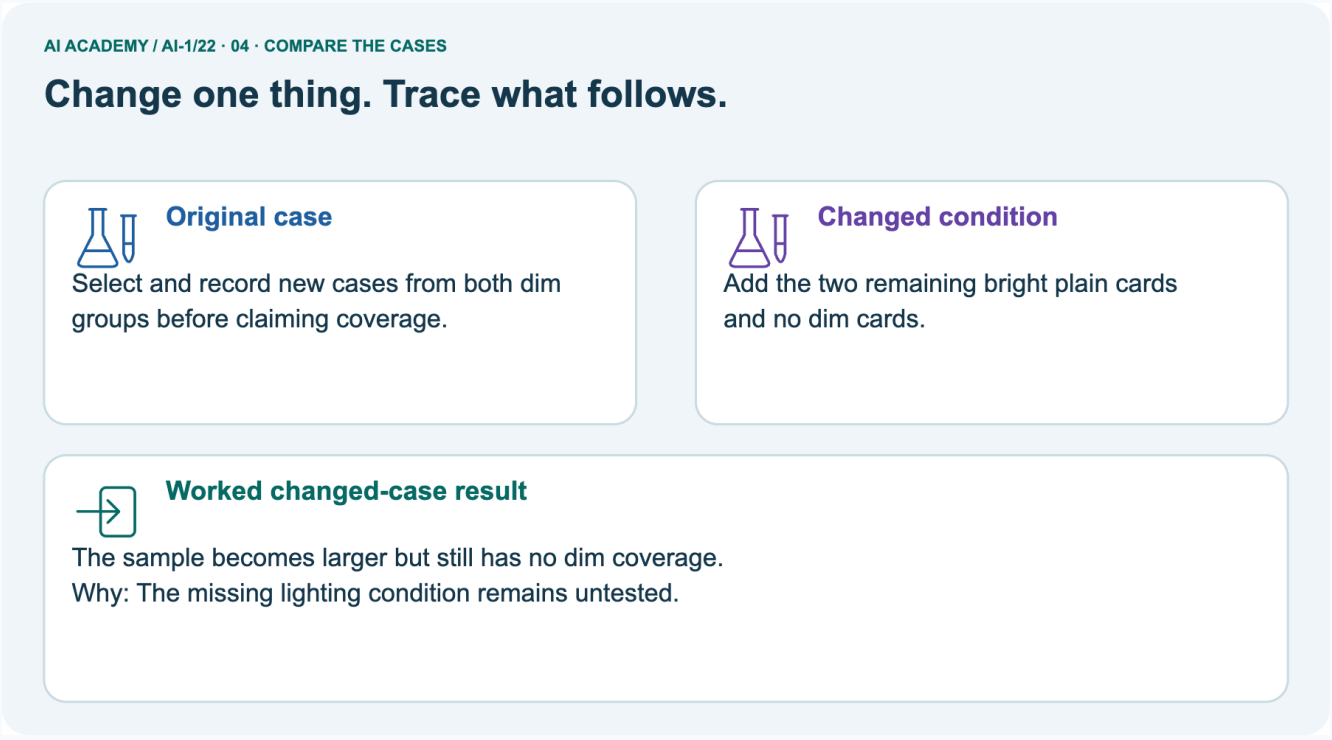
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Select and record new cases from both dim groups before claiming **coverage**.
Changed condition: Add the two remaining bright plain cards and no dim cards.

Expected result: The **sample** becomes larger but still has no dim **coverage**. Reason: The missing lighting condition remains untested.

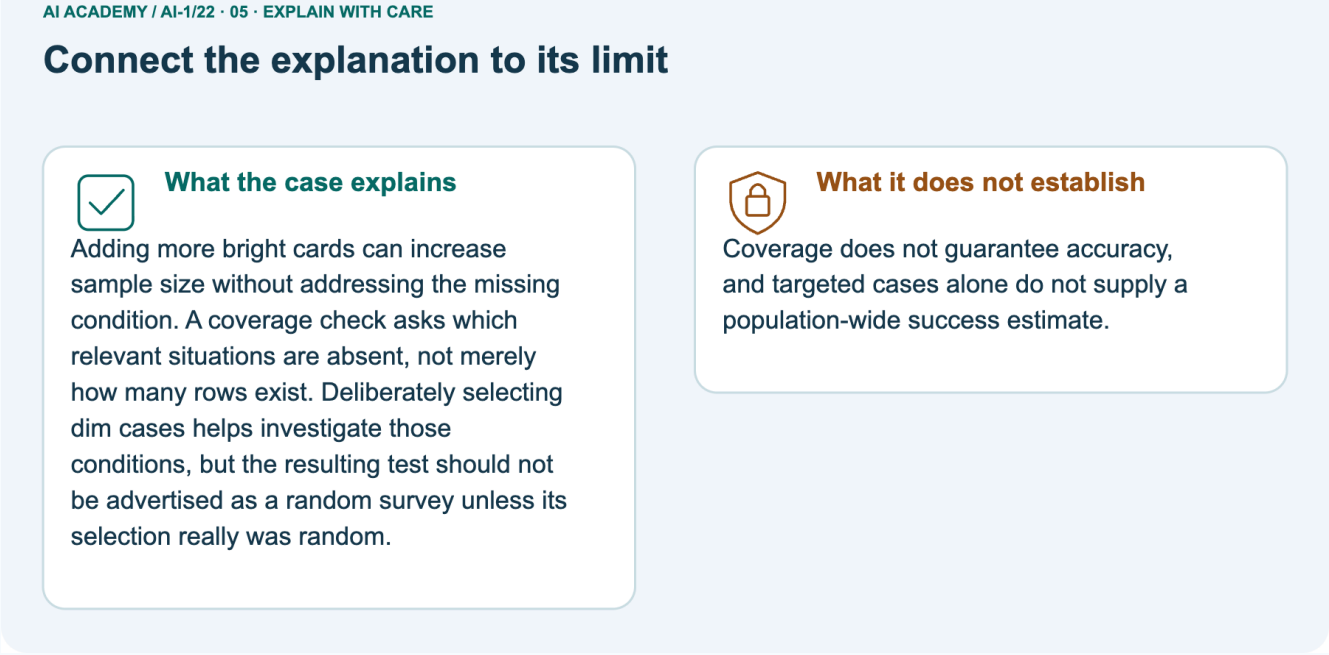
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The missing lighting condition remains untested.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

Limit: **Coverage** does not guarantee accuracy, and targeted cases alone do not supply a population-wide success estimate.

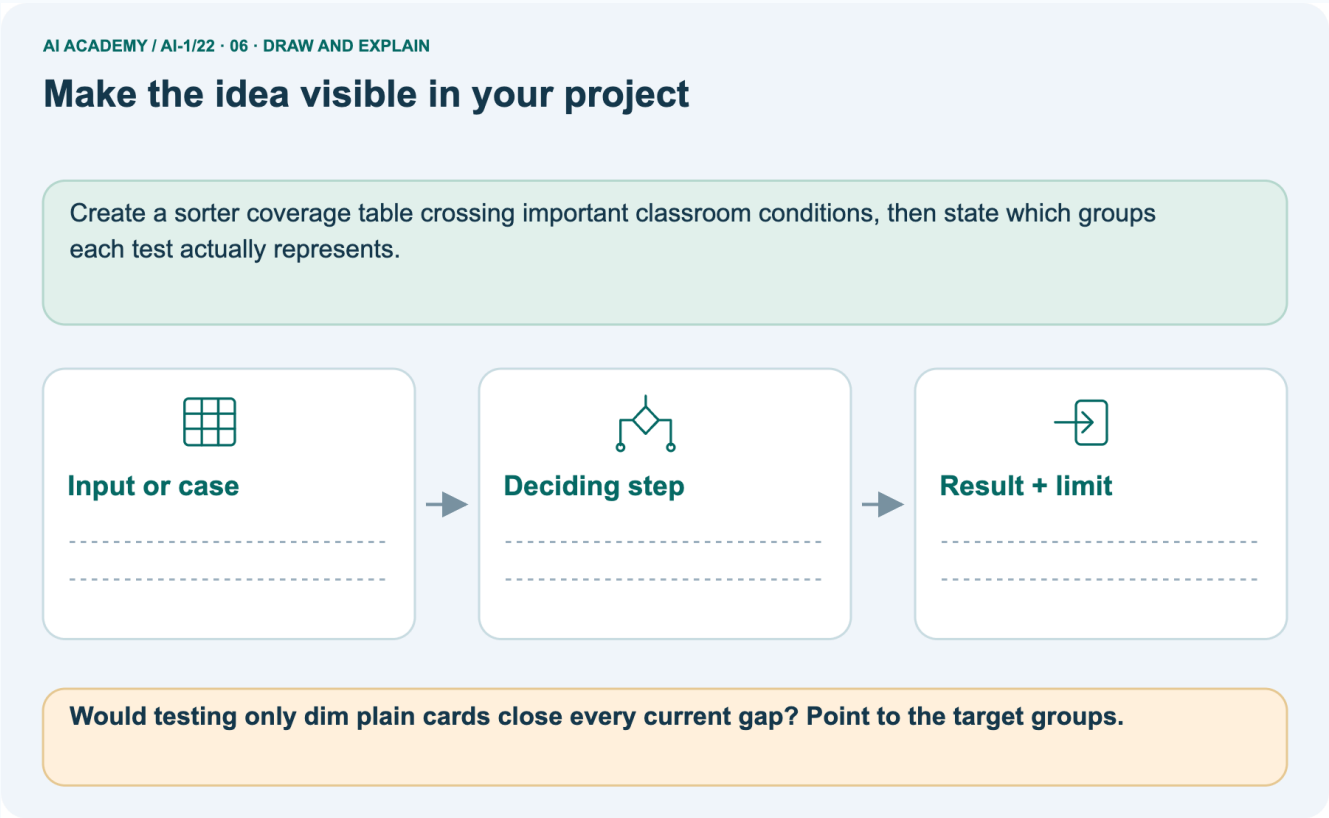
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Coverage** does not guarantee accuracy, and targeted cases alone do not supply a population-wide success estimate.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents. Draw your **input**, deciding step, result and limit.

Supported question: Would testing only dim plain cards close every current gap? Point to the target groups.

> Draw your project trace and justify the deciding step using **evidence**.

> **Discuss while showing the picture**
>
> Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: No. Dim patterned cards would still be absent; both dim groups need planned **coverage**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.